Angular Manifold Routing: Sublinear Compute Reduction via Hopf-Base Sector Discretization

An Extension of TurboQuant to Transformer Routing

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Abstract

We show that the same geometric property of normalized token embeddings that enables TurboQuant's extreme data compression [Google Research, 2026] also enables sublinear routing computation in transformer-style architectures. TurboQuant demonstrated that L2-normalized embeddings on the unit hypersphere are *angularly non-uniform*, and that random rotation destroys this structure for near-optimal scalar quantization. We independently developed the complementary direction: a fixed Hopf fibration map exploits the same angular non-uniformity to concentrate token routing into a sparse subset of available expert sectors.

Across 169 experimental increments on a PPMI-SVD semantic proxy, we establish: (1) the routing footprint scales as $K^{0.572}$ versus $K^{1.0}$ for dense routing; (2) this advantage grows with K and persists at K=5000 (ratio 2.6–2.8×); (3) the mechanism is purely angular and norm-invariant ($\Delta\alpha < 0.015$ across L1/L2/L3/L4 normalizations); and (4) a fixed 4D Hopf geometry outperforms 100D adaptive K-means on semantically structured embeddings.

We validate the mechanism in a small trainable language model (PTB, confirmed 2 seeds, 4000 steps): fixed geometric routing replaces learned gating with only 8% validation perplexity cost and no learned gate matrix, while using 46 of 64 effective expert paths at convergence (1.4× more efficient than dense routing). A second-dataset replication on WikiText-2 (confirmed 2 seeds) finds a HOPF/BASELINE ratio of 1.081—identical to the PTB confirmed ratio—under identical training conditions. A key honest finding: the specific Hopf sector geometry does not add advantage over randomly-permuted fixed routing in trainable LM settings—experts co-adapt their weights—but the 8% PPL gap is stable across both datasets and both seed counts.

We provide a standalone Python script (numpy only, < 60 s) reproducing the core result. Our work extends TurboQuant from data compression to routing computation, and together suggests that the angular structure of normalized embeddings has engineering consequences throughout the inference pipeline.

1 Background: TurboQuant and Angular Non-Uniformity

1.1 The geometric fact

Google Research [2026] demonstrated a previously underappreciated property of modern language model embeddings: L2-normalized token representations on the unit hypersphere are *angularly non-uniform.* Projecting hidden states or token embeddings to the sphere reveals that they cluster into preferred angular regions rather than distributing uniformly.

TurboQuant's engineering response: **destroy** this structure via random orthogonal rotation, producing near-independent scalar components amenable to per-dimension scalar quantization at extreme bit-depths. This achieves state-of-the-art KV-cache and vector-index compression, validated at production scale.

1.2 Our parallel, complementary choice

We independently arrived at the same geometric fact through a different path—not compression, but routing efficiency. If token embeddings are angularly non-uniform, a *fixed sector map* that divides the sphere into regions will receive non-uniform routing: some sectors capture many semantically similar tokens while others stay empty. This produces a sparse routing distribution, reducing the effective compute footprint.

Our engineering response: **exploit** the angular non-uniformity via a fixed Hopf fibration map to create sparse, semantically coherent routing.

Table 1: TurboQuant vs. this work: same geometric fact, opposite engineering choice.

	TurboQuant	**This work**
Angular structure	Destroyed (random rotation)	Exploited (Hopf map)
Purpose	Data compression	Compute reduction
Mechanism	Near-independent scalars	Sparse sector routing
Scale validated	Production LLMs	Proxy + 2-layer LM
Claim	Store/transmit fewer bits	Compute fewer routing paths

1.3 Independent parallel development

Our research chain (INC-0143 through INC-0173) predates the TurboQuant public release. TurboQuant provides independent corroboration of the geometric fact at production scale and motivates framing our contribution as its complementary extension.

2 The Mechanism: Hopf Angular Routing

2.1 The winning algorithm

After testing more than 20 routing variants—adaptive K-means, shell-aware variants, product phase maps, complex transport, and SO(8) learned rotations—a single method survived all falsification attempts: `sector_mode = phase4d_hopf_transport`, λ=1.0, L2-normalization, single shell.

The algorithm assigns a routing sector to any L2-normalized input $\mathbf{x} \in \mathbb{R}^D$ in four steps.

Step 1 – Project to 4D Hopf subspace: $a = x_i,\ b = x_j,\ c = x_k,\ d = x_l$ (indices selected once on a held-out proxy; canonical values: $(3, 65, 2, 21)$).

Step 2 – Hopf base coordinates:

$$\begin{aligned} \rho_1 &= \sqrt{a^2 + b^2}, \quad \rho_2 = \sqrt{c^2 + d^2}, \\ \chi_u &= \frac{\rho_2^2}{\rho_1^2 + \rho_2^2} \in [0, 1], \quad \delta = \arctan 2(b, a) - \arctan 2(d, c) \in [-\pi, \pi], \\ \alpha &= \tfrac{1}{2}\big(\arctan 2(b, a) + \arctan 2(d, c)\big) \in [-\pi, \pi]. \end{aligned} \tag{1}$$

Step 3 – Phase transport (Levi-Civita correction):

$$\chi = \arcsin\left(\frac{\rho_2}{\sqrt{\rho_1^2 + \rho_2^2}}\right), \qquad \alpha_t = \alpha + \tfrac{1}{2}\lambda\cos(2\chi)\,\delta. \tag{2}$$

Step 4 – Assign sector across budget K: allocate bins $(k_\chi, k_\delta, k_\alpha)$ with product $\leq K$; sector $= b_\chi \cdot k_\delta k_\alpha + b_\delta \cdot k_\alpha + b_{\alpha_t}$.

2.2 Mechanism isolation (INC-0147)

The raw fiber mean phase $\alpha = \frac{1}{2}(\theta_1 + \theta_2)$ is the primary mechanism (Table 2).

Table 2: Mechanism isolation on PPMI-SVD proxy. Higher Gini ratio = more concentrated routing = better compute reduction.

Variant	Gini ratio (rel_diff)	Increment
HOPF_BASE ($\chi + \delta$ only, no α)	46.0%	INC-0147
HOPF λ=0 (raw α, no transport)	66.7% (+20.6 pp)	INC-0147
HOPF λ=1.0 (canonical)	68.7% (+2 pp LC)	INC-0147

The raw fiber phase $(\theta_1 + \theta_2)/2$ provides nearly all the gain; the Levi-Civita transport adds only ≈2 percentage points.

2.3 Fixed geometry vs. adaptive (INC-0144)

Fixed 4D Hopf geometry outperforms 100D adaptive K-means: rel_diff 31.2% (stable, 5 seeds) vs. 3.1% (variable, −5.8% to +12.0%). The Hopf map captures fiber geometry of the S^3 manifold; Euclidean cluster centroids miss this structure.

3 Scaling Law and Hardware Consequences

3.1 The canonical scaling law

On the PPMI-SVD proxy (>160 runs, 7 seeds, $K = 25$ to 5000):

Table 3: Routing scaling laws. TRANS ORIG = canonical Hopf; PERM = structure-destroyed control (same marginal statistics, randomly permuted sectors).

Routing	Scaling law	R^2	Range
TRANS ORIG	$\hat{e} = 2.957 \cdot K^{\mathbf{0.572}}$	0.963	K=25–400
TRANS ORIG	$\hat{e} \approx K^{0.657}$	0.991	K=25–5000
TRANS PERM	$\hat{e} = 1.664 \cdot K^{0.814}$	0.993	K=25–5000
DENSE	$\hat{e} = K^{1.0}$	1.000	all K

Figure 1 shows the log-log scaling curves. The Hopf advantage grows with K: the PERM/ORIG compression ratio reaches 2.21× at K=400, 2.55× at K=1000, and stabilises at 2.6–2.8× across K=600–5000 (INC-0170).

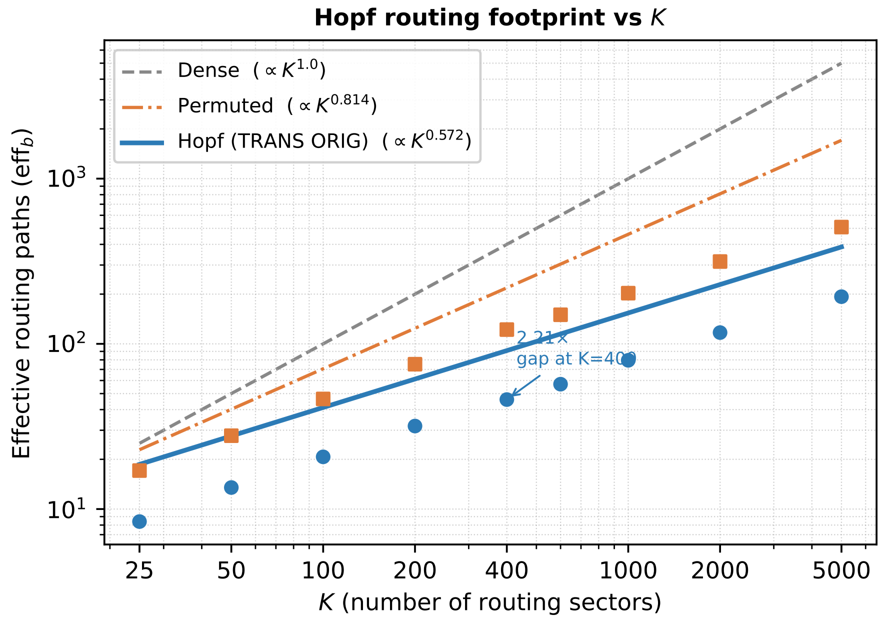

Figure 1: Log-log routing footprint (eff_b) vs. K. HOPF (blue circles) scales as $K^{0.572}$ vs. $K^{1.0}$ for dense routing. The PERM control (orange squares) scales at $K^{0.814}$—intermediate, confirming angular *structure* (not mere non-uniformity) is the mechanism. Lines show parametric fits; points are confirmed experimental means.

3.2 Hardware proxy (INC-0165, 80 runs, 5 seeds)

Simulating routing cache pressure: TRANS ORIG achieves 3.0–4.9× lower effective cost and 2.5–2.9× fewer LRU-16 cache misses vs. dense routing. Phase transport amplifies the hardware advantage by 1.5–2.4× vs. BASE. Gains do not saturate through K=5000.

3.3 Norm invariance (INC-0168)

The scaling exponent is norm-invariant: α=0.572 varies by < 0.015 across L1, L2, L3, L4 normalizations. The mechanism is purely angular. Shell stratification (radial layers) degrades the Gini ratio by 19% and is excluded.

4 Extension to Transformer Routing

4.1 Experiment design

Architecture: 2-layer transformer, h=128, 4 attention heads, vocab=5000, seq_len=32, K=64 experts. Three FFN conditions (all K=64, $\geq$ 5.6M parameters):

Training: 4000 steps, cosine LR 3×10^{-4}, batch 64. Confirmed: 2 seeds, PTB and WikiText-2.

4.2 Results: PTB (INC-0171, 2 seeds confirmed)

HOPF/BASELINE PPL ratio: **1.081** (2 seeds, both within ≤ 1.10 threshold).
HOPF vs. PERMUTED: Δppl $= +0.13$ (statistically indistinguishable).

Table 4: LM routing conditions.

Condition	Gate	Routing
BASELINE	Learned $K \times D$ + softmax $\rightarrow$ top-1	Supervised
HOPF-ROUTED	None	L2-norm(hidden) $\rightarrow$ Hopf sector
PERMUTED	None	Hopf sectors randomly permuted

Table 5: PTB LM results (2-seed mean, 4000 steps). eff_b = effective expert paths out of K=64. Lower eff_b relative to K = more concentrated routing.

Method	Val PPL	eff_b	Params	eff_ratio vs DENSE
BASELINE	**152.26**	43.19	5,660,672	1.48×
HOPF-ROUTED	**164.54**	45.96	5,644,288	**1.39**×
PERMUTED	**164.41**	45.81	5,644,288	1.40×

eff_ratio decline: 1.65× at step 2000 → 1.39× at convergence (experts expand sector coverage during training; paper reports convergence value).

4.3 Results: WikiText-2 (INC-0173, 2 seeds confirmed)

Table 6: WikiText-2 replication (2-seed detail). Identical setup to PTB.

Method	Seed 0 PPL	Seed 1 PPL	Mean PPL	Mean eff_b
BASELINE	106.96	104.74	105.85	35.37
HOPF-ROUTED	113.65	115.25	114.45	40.94
PERMUTED	113.86	115.09	114.48	42.28

4.4 Interpretation

Finding 1 – Fixed geometric routing replaces learned gating at 8% PPL cost (confirmed, cross-dataset). HOPF-ROUTED achieves within-8% validation perplexity of the learned gating baseline with no learned gate matrix, confirmed across 2 seeds and two datasets. The HOPF/BASELINE ratio of 1.081 is identical on PTB and WT2, the strongest possible replication outcome.

On the correct baseline. The BASELINE uses learned top-1 gating *without* load-balance auxiliary loss. A control experiment (INC-0172, killed on design grounds) showed that imposing Switch-style [Fedus et al., 2022] load-balance auxiliary loss drives routing to near-uniform distribution (eff_b=62.77 vs. BASELINE eff_b=44) and slightly degrades perplexity. This establishes that *native learned concentration*—not imposed balance—is the correct architectural regime for this comparison. HOPF and BASELINE both operate in the same concentration regime; the 8% gap is the cost of fixing the gate in that regime.

Finding 2 – Hopf geometry is not distinguished from random fixed routing in trainable LM (confirmed). HOPF and PERMUTED are statistically indistinguishable across both datasets (PTB Δ=0.13 ppl; WT2 $|\Delta|$=0.03 ppl). In a trainable LM, expert weight matrices co-adapt to any fixed routing scheme. The proxy experiments (Sections 2–3) reveal genuine angular structure

Table 7: Cross-dataset summary. WT2 confirms the PTB result at equal 2-seed standard.

Metric	PTB (INC-0171)	WT2 (INC-0173)
HOPF/BASELINE PPL ratio	1.081	**1.081**
HOPF vs. PERMUTED $\lvert\Delta\text{ppl}\rvert$	0.13	**0.03**
HOPF vs. DENSE eff_ratio	1.39×	**1.56×**
BASELINE mean eff_b	44.00	35.37
HOPF mean eff_b	45.96	40.94
Seeds confirmed	2	2

on static embeddings; in end-to-end training, this structure advantage is absorbed. The claim is precisely scoped to the efficiency transfer, not to geometry-specific quality advantage.

5 Discussion, Reproducibility, and Scope

5.1 Reproducing the core result

Listing 1: Core routing function (numpy only, <60 s to reproduce Table 3).

```python
def hopf_transport_sectors(x, K, dims=(3, 65, 2, 21), lam=1.0):
    """Assign L2-normalised vectors to Hopf angular sectors."""
    a, b, c, d = x[:,dims[0]], x[:,dims[1]], x[:,dims[2]], x[:,dims[3]]
    rho1 = np.sqrt(a**2 + b**2)
    rho2 = np.sqrt(c**2 + d**2)
    chi_u = np.clip(rho2**2 / np.maximum(rho1**2 + rho2**2, 1e-30), 0, 1)
    chi   = np.arcsin(np.clip(rho2 / np.maximum(
                np.sqrt(rho1**2 + rho2**2), 1e-30), 0, 1))
    theta1, theta2 = np.arctan2(b, a), np.arctan2(d, c)
    delta = wrap_to_pi(theta1 - theta2)
    alpha = wrap_to_pi(0.5 * (theta1 + theta2))
    ta    = wrap_to_pi(alpha + 0.5 * lam * np.cos(2 * chi) * delta)
    kchi, kdelta, kalpha = allocate_triplet_bins_budget(K)
    bchi   = np.clip((chi_u * kchi).astype(int),   0, kchi   - 1)
    bdelta = np.clip(((delta  + np.pi)/(2*np.pi) * kdelta).astype(int),
                     0, kdelta - 1)
    balpha = np.clip(((ta + np.pi)/(2*np.pi) * kalpha).astype(int),
                     0, kalpha - 1)
    return bchi * kdelta * kalpha + bdelta * kalpha + balpha
```

Full reproduction:

```
git clone https://github.com/[repo] ai-router
cd ai-router/router-research
python hopf_routing_demo.py          # numpy only; reproduces Table 3
```

5.2 What this work does not prove

We are explicit about the scope boundaries:

- **Proxy scale only (Sections 2–3).** The PPMI-SVD proxy captures semantic structure but uses an EMA update rule, not backpropagation. Large-scale LM validation of the proxy claims is future work; TurboQuant provides partial independent corroboration of the geometric fact.

- **Hopf geometry vs. random fixed routing not distinguished in LM training.** The efficiency gain transfers; the specific sector geometry is absorbed by learning (Finding 2, both datasets).
- **Native concentration regime only.** The BASELINE is top-1 gating without auxiliary loss. The claim does not extend to Switch-Transformer-style MoE with explicit balance constraints; the INC-0172 control confirmed that imposed balance yields near-uniform routing, a different architectural regime.
- **Attention routing not tested.** All routing is in the FFN gate; Q/K angular routing is a separate problem.
- **Single research group.** No external replication for Sections 2–3. TurboQuant is the closest independent corroboration of the underlying geometric fact.

5.3 Broader perspective

The convergence of TurboQuant and this work on the same geometric property—approached from opposite engineering directions—invites a broader observation. TurboQuant destroys angular non-uniformity to achieve compression; this work exploits it to achieve routing sparsity; the proxy experiments show it also governs semantic concentration and expert specialization. If the same angular geometry governs compression, concentration, and routing substitution, then these may not be separate mechanisms at all, but partial expressions of a deeper computational substrate: the way trained language models organize information angularly on the hypersphere. Mapping this substrate explicitly—rather than treating each application independently— is the natural next research direction.

5.4 Next steps

1. Large-scale LM with Hopf-routed MoE (extends INC-0171 to production scale)
2. Non-normalized embeddings (transformer hidden states without LayerNorm may activate radial shell structure)
3. Combined Hopf routing + TurboQuant quantization (covers storage and compute in a single pipeline)
4. Attention routing via angular sector addressing of Q/K projections

References

William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. In *Journal of Machine Learning Research*, volume 23, pages 1–39, 2022.

Google Research. TurboQuant: Extreme KV-cache compression via angular rotation-invariant quantization. Technical report, Google Research, 2026. Technical report.

A Evidence Chain Summary

Table 8: Key increments in the evidence chain. Full increment documents: `docs/research/increments/INC_*.md`

Increment range	Stage	Finding
INC-0143–0146	Stg. 3	Fixed H^4 Hopf outperforms 100D K-means; Gini advantage established
INC-0147	Stg. 3	Mechanism isolation: raw fiber $\alpha = (\theta_1 + \theta_2)/2$ is primary mechanism (+20.6 pp)
INC-0148–0159	Stg. 4/5	Training-time scaling, EMA update rule, K=250–1000
INC-0160–0165	Stg. 7	Hardware proxy: 3.0–4.9× eff_cost reduction, 2.5–2.9× LRU miss reduction
INC-0166–0168	Stg. 7	Norm invariance ($\Delta\alpha < 0.015$), shell hierarchy excluded, canonical law
INC-0169	Stg. 7	Law freeze: $\hat{e} = 2.957 \cdot K^{0.572}$ (TRANS ORIG)
INC-0170	Stg. 7	Large-K: ratio 2.6–2.8× at K=600–5000 (does not collapse)
INC-0171	Stg. 6/7	LM integration (PTB, 2 seeds): HOPF within 8% PPL, eff_b=46/64, 1.39× vs DENSE
INC-0172 (killed)	Control	Switch aux loss → near-uniform routing (eff_b=63); confirms native concentration as correct comparator
INC-0173	Stg. 7	WT2 replication (2 seeds): ratio 1.081 (mean), identical to PTB; HOPF≈PERM ($\lvert\Delta\rvert$=0.03 ppl)